cpts 317. 11/4

machine $\longrightarrow$ Grammar.

PDA c.f. grammar.

Alg: ① $\delta(p, a, A) \ni (q, \lambda)$ // a PDA instruction // pop the stack.

$$\underbrace{[p, A, q]}_{\text{one nonterminal symbol}} \longrightarrow a$$ is a grammar rule

② $\delta(p, a, A) \ni (p_1, \gamma)$ with $\gamma \neq \Lambda$.

let $\gamma = B_1 \cdots B_m$

$$[p, A, q] \longrightarrow a[p_1, B_1, p_2] \cdots [p_{m-1}, B_m, p_m]$$

with $p_m = q$.

③. Initial rule in G:

$$S \longrightarrow [q_0, Z_0, q]$$

Exaple. Given a PDA with

input alphabet

stack alphabet

$$Q = \{q_0, q_1, q_2\}, \quad \Sigma = \{0, 1\}, \quad \Gamma = \{0, 1, Z_0\}$$

$$A = \{q_2\}.$$ $\longrightarrow$ this doesn't matter since we use empty stack acceptance.

Instructions (code):

$$\delta(q_0, 0, Z_0) = \{(q_0, 0Z_0)\}$$

$$\delta(q_0, 0, 0) = \{(q_0, 00)\}$$

$$\delta(q_0, 1, 0) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, 1, 0) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_2, \Lambda)\}.$$ empty stack acceptance.

Translate the code into grammar:

For: $\delta(q_0, 1, 0) \ni (q_1, \Lambda)$, I have:

$$[q_0, 0, q_1] \longrightarrow 1$$

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For: $\delta(q_0, 0, Z_0) \ni (q_0, 0Z_0)$, I have:

$$[q_0, Z_0, *_1] \longrightarrow 0\,[q_0, 0, *_2]\,[*_2, Z_0, *_1],$$

where $*_1, *_2 \in Q$.

For: $\delta(q_0, 0, 0) \ni (q_0, 00)$

$$[q_0, 0, *_1] \longrightarrow 0[q_0, 0, *_2][*_2, 0, *_1]$$

where $*_1, *_2 \in Q$.

For $\delta(q_1, \Lambda, Z_0) \ni (q_2, \Lambda)$, I have

$$[q_1, Z_0, q_2] \longrightarrow \Lambda$$

Additionally, I have

$$S \longrightarrow [q_0, Z_0, *_1] \quad \text{where } *_1 \in Q.$$

For the grammar obtained, how many rules?

$1 + 1 + 9 + 9 + 1 + 3 = ?$

Summary:

1. c.f. grammar.

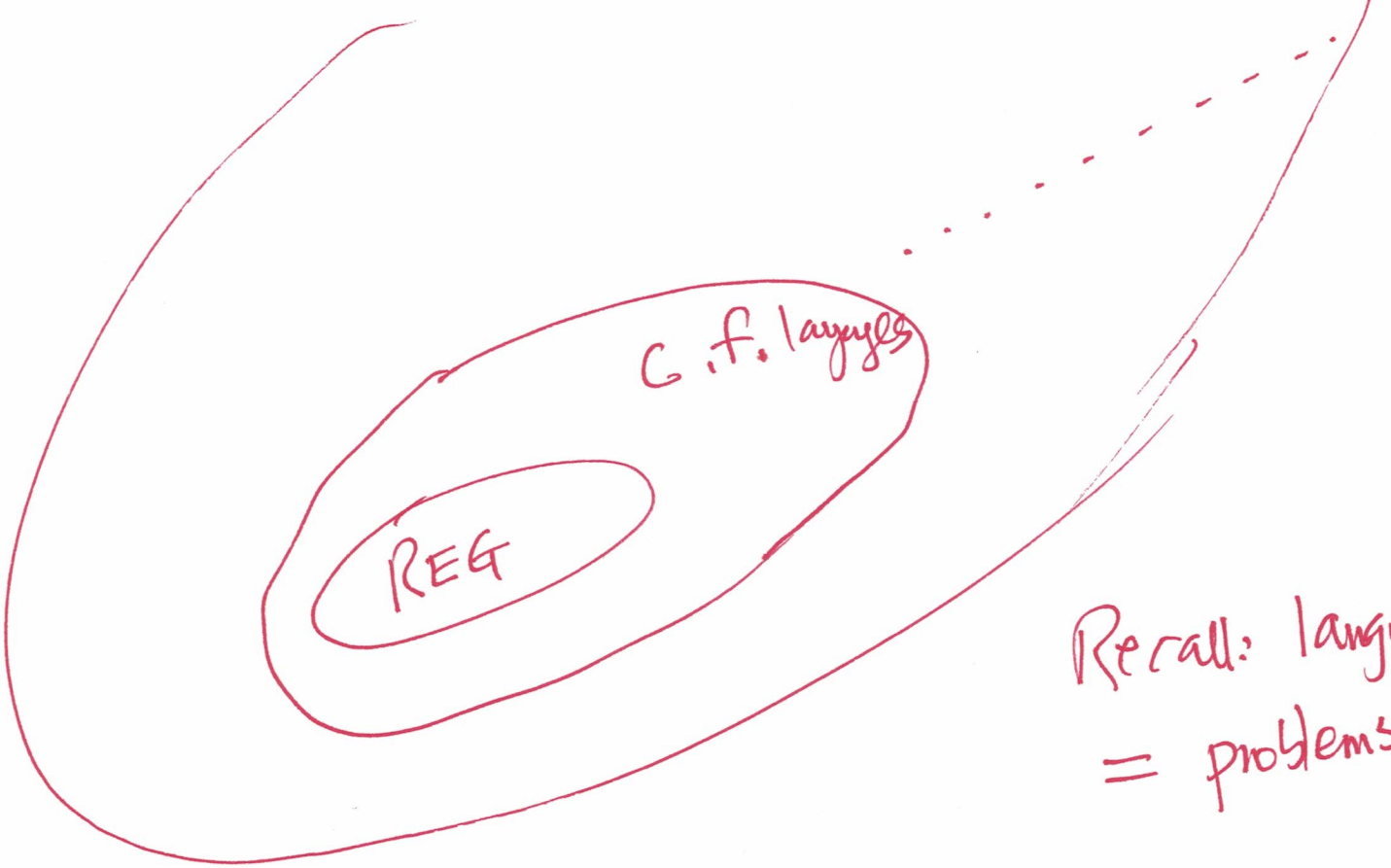
2. PDA

3. languges accepted by PDA are exactly c.f. languges ; $\begin{cases} 1. \text{ Grammar} \to \text{machine} \\ 2. \text{ Machine} \to \text{Grammar} \end{cases}$

Recall: For regular languges, we had Pumping Lemma.

use PL to show: a language is not reg.

How about c.f. languges ?



Recall: languges
= problems.

Pumping Lemma for c.f. languages.

Theorem. Let L be an infinite c.f. language. Then there is a positive integer n s.t. for each word $z \in L$ with $|z| \geq n$, z can be written into 5 parts

$$z = uvwxy$$

for some u, v, w, x, y s.t.:

1. $|vwx| \leq n$
2. $|vx| \geq 1$
3. $\forall i \geq 0$, $uv^iwx^iy \in L$.

L is given as a c.f. language.

$\Rightarrow$ $\exists\, n > 0$: // Never assume $n = 5$.

$\forall$ long word $z \in L$, // $|z| \geq n$.

$\exists\, u, v, w, x, y$ with $z = uvwxy$:

① $|vwx| \leq n$

② $|vx| \geq 1$

③ $\forall i \geq 0$, $uv^iwx^iy \in L$.

// most CS students are going to mess up the logic.